Is Medical Pretraining enough when the modality is different?

A study focused on Polyp Segmentation

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Abstract

Transfer learning is essential in medical imaging due to limited labeled data. While ImageNet is commonly used for pretraining, domain-specific alternatives like RadImageNet show promise in radiology tasks. However, their effectiveness in other modalities, such as endoscopy, is less understood. We evaluate the impact of pretraining on ImageNet, RadImageNet, and a histopathology dataset for endoscopic polyp segmentation. Using ResNet-50 and ViT-Small backbones with a DeepLabV3+ decoder, we fine-tune on CVC-ClinicDB, Kvasir-SEG, and SUN-SEG. Results show ImageNet pretraining consistently outperforms medical-domain pretraining, underscoring the importance of modality alignment in transfer learning for medical imaging.

Keywords- Polyp segmentation, DeepLabV3+, ViT, ResNet, Transfer learning

Introduction

Medical images differ significantly from natural images in both semantics and acquisition methods. Moreover, medical imaging spans diverse modalities, such as radiology, histopathology, and endoscopy, each with distinct characteristics. Endoscopy is particularly interesting as it involves camera-based capture like natural images but exhibits unique semantic features. The effectiveness of medical image like radiology or histopathology pretraining used in endoscopy imaging tasks remains unclear.

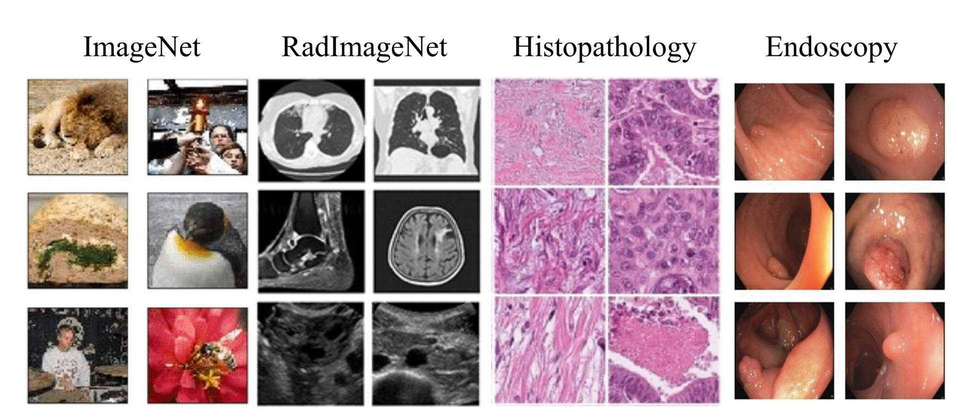

Figure 1: Natural, radiology, histopathology and endoscopy image samples.

Workflow

Figure 2 depicts the steps in the experiments. ResNet50 and Vit-Small are pretrained on three datasets, and finetuning is performed on three public polyp segmentation datasets.

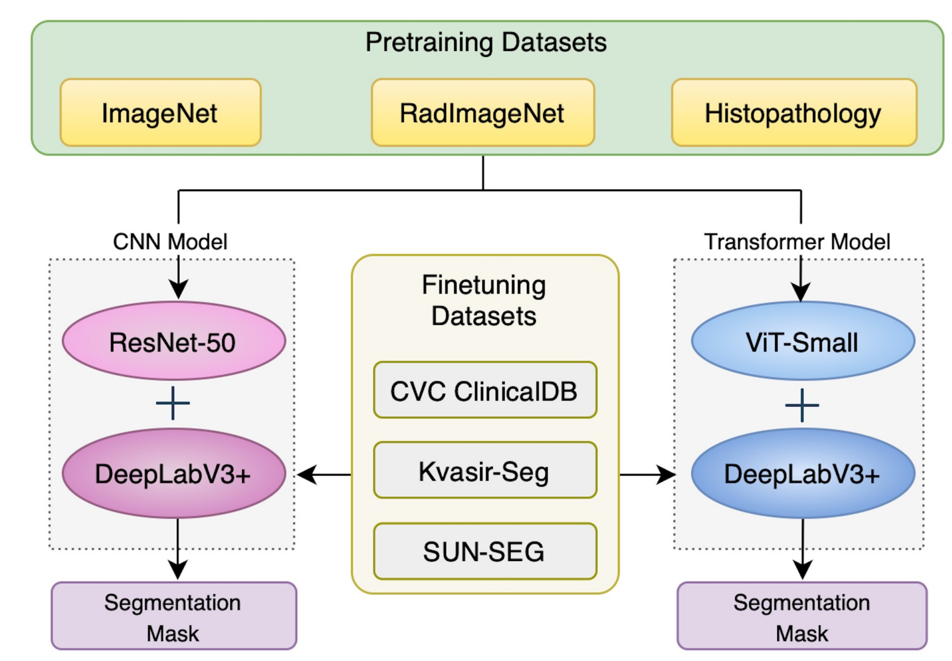

Figure 2: Pipeline of the proposed method.

Datasets

Pretraining Datasets: The datasets used for pretraining can be broadly categorized as comprising of either natural or medical images.

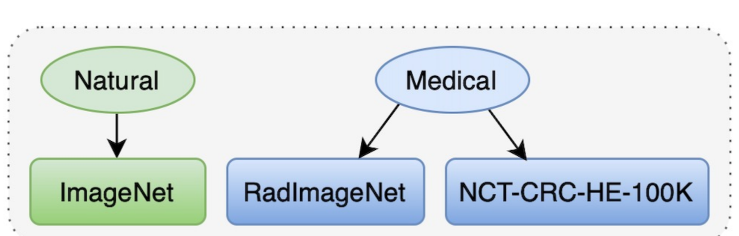

Figure 3: Datasets used for pretraining ResNet50 and ViT-Small

Finetuning Datasets: We evaluate our pretrained models on three publicly available polyp segmentation datasets.

Figure 4: Polyp segmentation datasets used for finetuning.

Method

- An encoder-decoder framework is followed, with ResNet-50 or ViT-Small as the encoder and a DeepLabV3+ decoder to generate the segmentation output.
- Input images are resized to 224×224 and passed through the encoder to extract features.
- For ResNet-50, the final convolutional layer output is fed into an Atrous Spatial Pyramid Pooling (ASPP) module to capture multi-scale context, while a low-level feature map from layer1 is fed in a skip connection to the decoder.
- For ViT-Small, intermediate feature maps are obtained. The final transformer block output is processed by ASPP, and the earliest feature map is used as the decoder skip connection.
- The decoder combines ASPP output with low-level features and produces a segmentation mask.

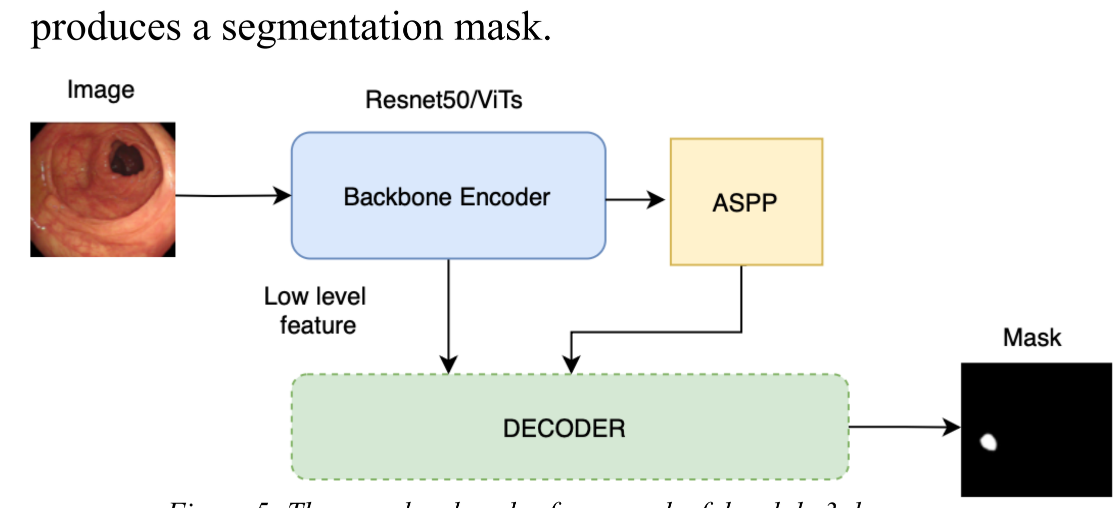

Figure 5: The encoder-decoder framework of deeplabv3plus.

The two backbone models of our choice were Resnet50 and ViT-Small. These models were chosen because they have similar or comparable number of parameters.

Experiments & Results

We evaluate our pretrained models on three publicly available polyp segmentation datasets. The following experimental settings are consistent across both models and all fine-tuning datasets:

- **Loss Function:** Binary Cross-Entropy
- **Optimizer**: Adam with a learning rate of 0.0001
- **Training Duration:** 20 epochs
- **Batch Size**: 32
- **Evaluation Metrics:** Dice Coefficient and Intersection over Union (IoU)
- **Model Selection:** Best-performing model
- **Hardware:** 4 × NVIDIA TITAN RTX GPUs were used for both training and inference

Table 1: Dice and IoU scores of the finetuned models- ResNet50 and ViT-Small, pretrained on ImageNet (IN), RadImageNet (RIN) and NCT-CRC-HE-100K (Histopathology) datasets.

Model	Dataset	Dice			IoU		
	Metrics	IN	RIN	HP	IN	RIN	HP
ResNet50	CVC-DB	**0.8230**	0.5820	0.7262	**0.7389**	0.4634	0.6131
	Kvasir-SEG	**0.8244**	0.5279	0.7062	**0.7340**	0.4006	0.5947
	SUN-SEG	**0.9353**	0.9209	0.9307	**0.8862**	0.8657	0.8813
ViT-Small	CVC-DB	**0.8705**	0.5033	0.5520	**0.7913**	0.4024	0.4325
	Kvasir-SEG	**0.8706**	0.5228	0.5065	**0.7986**	0.3954	0.3839
	SUN-SEG	**0.9000**	0.8411	0.8453	**0.8334**	0.7580	0.7658

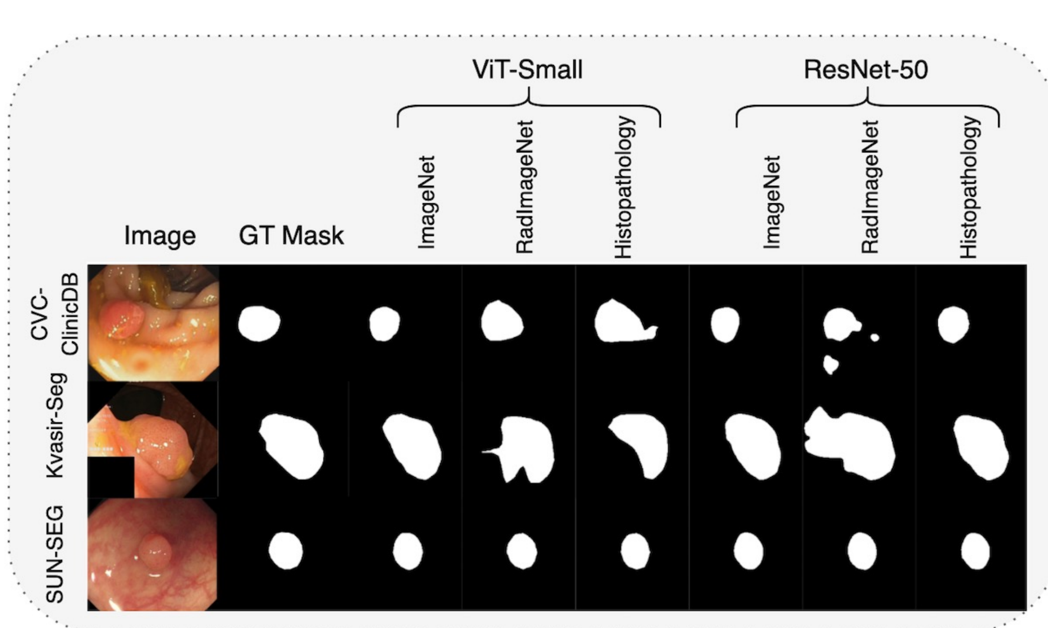

Figure 6: Samples of predicted masks from finetuned ViT-small and ResNet50 models.

- ImageNet-pretrained models consistently outperform RadImageNet or NCT-CRC-HE-100K pretrained ones, with the largest differences on smaller datasets.
- This pattern holds for both CNN and transformer architectures. The performance gap narrows on SUN-SEG, the largest dataset in our evaluation.
- Models pretrained on histopathology data perform comparably to those pretrained on ImageNet, with only slightly lower Dice and IoU scores, whereas RadImageNet pretrained models continue to underperform.
- This may be attributed to the similarity in color characteristics between histopathology and endoscopic images, in contrast to grayscale radiology images.

Discussion & Conclusion

- In this work, we conduct focused and controlled experiments to investigate how different medical modalities vs natural image pretrained models influence the effectiveness for polyp segmentation task.
- Our findings suggest that although larger finetuning datasets can reduce the impact of pretraining misalignment, features learned from ImageNet remain more transferable to non-radiological modalities such as endoscopy.

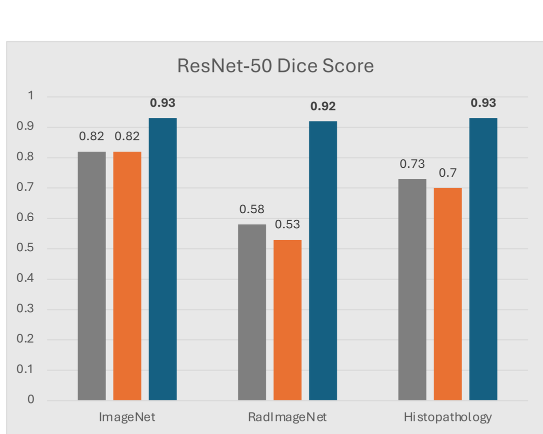

Figure 7(a): Dice scores of finetuned ResNet-50 models.

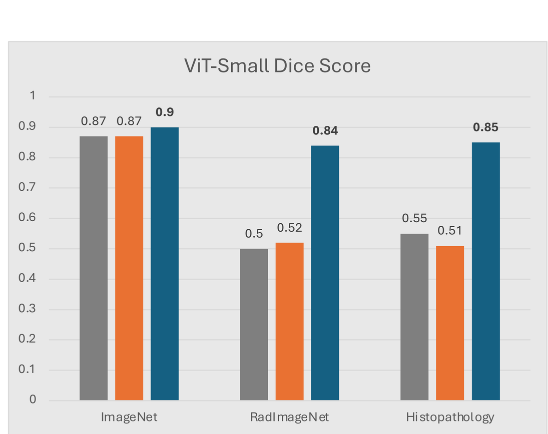

Figure 7(b): Dice scores of finetuned ViT-Small models.

Overall, our results indicate that medical-domain pretraining is not universally advantageous, and that modality-specific characteristics should guide the selection of pretrained models in medical imaging tasks.